SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0714

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

QUESTIONS: 5 **Total Marks:50**

Annexure A

Debugging & Optimisation Task

Code of Conduct:

I ***Y.Geetha Reddy(192572053)****(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Geetha

Annexure A

Debugging & Optimisation Task

2. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

<u>Task</u>

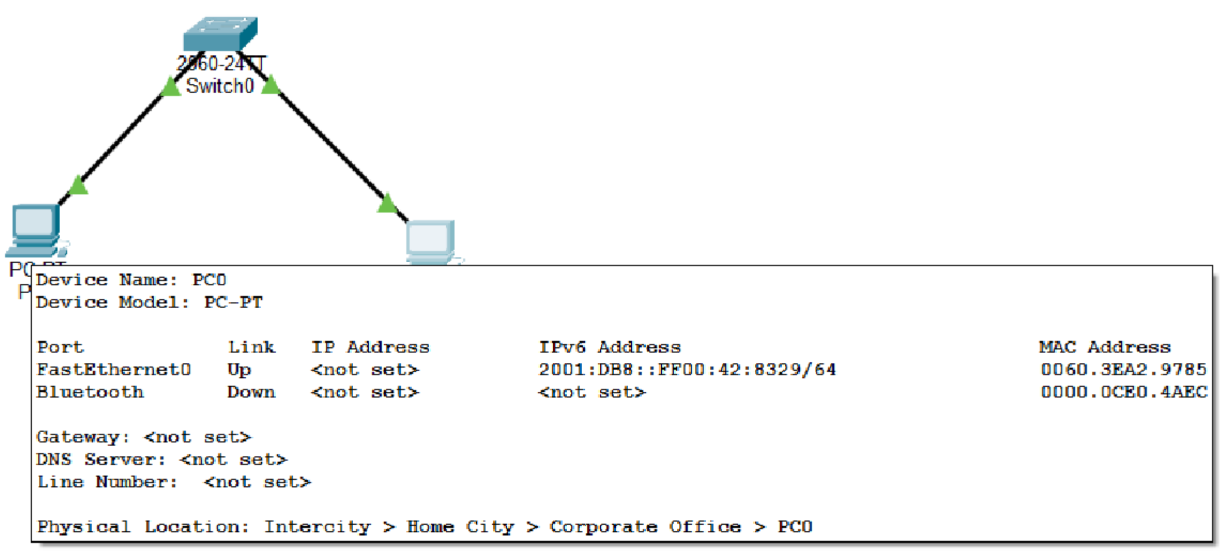
Analyze and optimize the subnet allocation by:

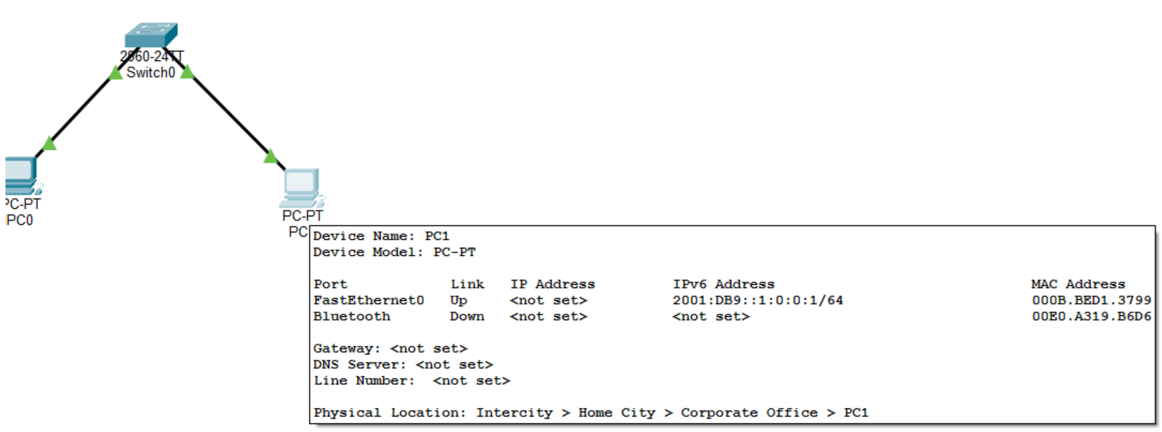
- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

ANSWER:

Initial Configuration

Device	IPv6 Address	Prefix Length
PC0	2001:db8::ff00:42:8329	/64





Expanded IPv6 Addresses
Device A
Compressed Address:2001:db8::ff00:42:8329
Expanded Address:2001:0db8:0000:0000:0000:ff00:0042:8329

Device B
Compressed Address:2001:db8:0:0:1::1
Expanded Address: 2001:0db8:0000:0000:0001:0000:0000:0001

Network Prefix Calculation
Since the prefix length is **/64**, the first four hextets represent the network portion.
Device A
2001:db8:0:0::/64
Device B
2001:db8:0:0::/64
Initially, both devices belong to the same subnet.

Debugging Scenario
To demonstrate IPv6 troubleshooting, the IPv6 address of PC1 was intentionally modified.
Modified IPv6 Address of PC1
2001:db9:0:0:1::1/64
Now,
PC0 Network
2001:db8:0:0::/64
PC1 Network
2001:db9:0:0::/64
Since the network prefixes are different, both devices belong to different subnets, causing communication failure.

```
C:\>ping 2001:db9:0:0:1::1

Pinging 2001:db9:0:0:1::1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 2001:DB9::1:0:0:1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Optimized IPv6 Configuration
After debugging, the IPv6 configuration was corrected.

Device	**Correct IPv6 Address**	**Prefix**
PC0	2001:db8::ff00:42:8329	/64
PC1	2001:db8:0:0::20	/64

Both devices now belong to the same subnet.

AFTER DEBUGGING:
Reply from 2001:DB8::20

Packets: Sent = 4
Received = 4
Lost = 0
Communication was successfully restored after assigning the correct IPv6 address.

Available Host Addresses
Prefix Length:/64
Remaining Host Bits:128 − 64 = 64
Available Host Addresses:2^64

= 18,446,744,073,709,551,616
Approximately 18.4 Quintillion host addresses are available in a /64 IPv6 subnet.

```
C:\>ping 2001:db8:0:0:1::1

Pinging 2001:db8:0:0:1::1 with 32 bytes of data:

Reply from 2001:DB8::1:0:0:1: bytes=32 time=10ms TTL=128
Reply from 2001:DB8::1:0:0:1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8::1:0:0:1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8::1:0:0:1: bytes=32 time<1ms TTL=128

Ping statistics for 2001:DB8::1:0:0:1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>
```

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation